

FROM CIFAR CLASSIFIERS TO CLIP ENCODERS

Invariance is not robustness

What actually predicts a model's adversarial robustness

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A model that ignores small shifts of its input is called invariant. A model that does not flip under a tiny worst-case change is called robust. We show these are different, measure what tells them apart, and correct a common mistake.

The puzzle

Does making a vision model more shift-invariant help or hurt its adversarial robustness?

One literature says: it helps.

Anti-aliasing and equivariant designs report better robustness. Smoother, more invariant models look safer.

Another says: it hurts.

Perfect shift-invariance can shrink the margin between classes, which should make attacks easier.

Both can't be right as stated. Which is it, and what is the real signal?

Takeaway. We resolve this with one simple quantity, and find that the field's usual invariance metric is the wrong thing to look at.

Two different things, often confused

Invariance

Ignoring changes that should not matter: a small shift, a crop, a reworded prompt.

Measured by *shift-consistency*: how often the prediction stays the same when you shift the image.

Adversarial robustness

Not being flipped by a small *worst-case* change chosen by an attacker.

Measured by running a strong attack (AutoAttack) and seeing how big a perturbation it takes to change the answer.

They sound related, and the field often uses the first as a proxy for the second. Our first finding is that this proxy does not work.

The measuring stick: a margin-to-sensitivity ratio

η/L , computed on clean images, no attack needed

$$\eta/L = \frac{\text{margin}}{\text{sensitivity}} = \frac{\text{how confident the model is}}{\text{how fast that confidence changes if you nudge the input}}$$

- The **margin** is the gap between the top class and the runner-up.
- The **sensitivity** is the size of the input gradient of that margin, measured in the way the attacker moves (the norm matched to the threat).
- A large ratio means: far from the decision boundary, and slow to get there. That is exactly what robustness should be.

Takeaway. η/L is a first-order estimate of how big a perturbation it takes to flip the label. It needs no attack and is not changed by rescaling the model's outputs.

Finding 1: the invariance metric is saturated

Shift-consistency looks the same for every modern model, robust or not

	shift-consistency	spread across models
16 frozen CLIP encoders (robust and non-robust alike)	0.96 – 0.99	almost none

- Every modern encoder is already near-perfectly shift-consistent, so the metric cannot tell them apart on *any* kind of robustness.
- Once you account for clean accuracy, it does not predict robustness to common corruptions either.
- The same squeeze appears on strong CNN backbones: consistency stops discriminating.

Takeaway. Shift-consistency is treated as a robustness signal, but it is saturated. It has no room left to rank models. This is why the “invariance predicts robustness” story keeps giving mixed answers.

Finding 2: “invariance helps” is a weak-attack artifact

The effect shows up under a cheap attack and vanishes under a strong one

non-robust encoder	weak attack (FGSM)	strong attack (AutoAttack)
robust accuracy	16% – 61%	0%

- Under a single-step FGSM attack, more invariant models look more robust: the “invariance helps” ordering appears.
- Under AutoAttack, every non-robust encoder drops to zero. There is no robustness left to order, and the correlation disappears.
- This is a classic gradient-masking signature: a weak attack overstates robustness.

Takeaway. Recent claims that architectural invariance gives robustness without adversarial training are measured with weak attacks. They do not survive a strong one.

Finding 3: the margin ratio predicts robustness

η/L orders robustness where invariance does not, across three settings

setting	η/L vs robustness	shift-consistency vs robustness
CIFAR-10 classifiers	+0.99	uninformative
ResNet on ImageNet-100	+0.90	anti-correlated
Frozen CLIP encoders (per image)	+0.74 to +0.96	near zero

- Same law from small CNNs to a ResNet to frozen foundation-model encoders.
- Within a single robust encoder, image by image, η/L tracks how large an attack it takes to flip each image, while shift-consistency does not.

Takeaway. The right question is not “how invariant is the model” but “how large is the margin relative to the sensitivity”. That is η/L .

Why it works: a two-sided bracket

The ratio is not just a heuristic; it brackets the true robust radius

$$\frac{\eta}{L} \leq \underbrace{r}_{\text{true robust radius}} \leq \frac{\eta}{\alpha}$$

- The **lower bound** is the standard certificate: the model cannot be flipped by anything smaller than η/L .
- The **upper bound** is the matching side: with α a slope along the path to the boundary, no perturbation larger than η/α can be safe.
- The gap between the two is a condition number $\kappa = L/\alpha$. When it is close to one, η/L is not just a bound, it is the answer.

Takeaway. η/L is the small-curvature limit of the true radius. This tells us exactly when it is tight, and points to what it misses.

Finding 4: detecting robustness is not the same as ranking it

η/L separates robust from fragile, but cannot order the robust ones

- Across all encoders, η/L cleanly **separates** the robust ones from the fragile ones.
- But *among the already-robust* encoders, η/L no longer orders them. Two encoders with the same η/L can differ in robustness.
- Something beyond the margin and the size of the gradient is at play.

So what ranks the robust models?

Takeaway. η/L uses the *size* of the gradient. To rank robust models we need its *shape*.

Finding 4: the shape of the gradient ranks robust models

A concentrated gradient resists an all-pixel attack better than a spread one

- An ℓ_∞ attack pushes on *every* pixel at once.
- If the margin's gradient is **concentrated** on a few pixels, most of that budget is wasted, and the model holds.
- If it is **spread** over many pixels, the attack engages all of them, and the model gives way.
- Measured by the gradient's **anisotropy** (how spread it is), this orders the robust encoders where η/L could not.

We can prove it in a clean case: when the model's curvature lines up with its gradient, a more spread gradient provably shrinks the robust radius, and the shrinkage grows with the spread. The direction of the effect we see in real encoders matches this case.

Takeaway. Robustness among strong models is a matter of *where* the model is sensitive, not only

Finding 5: adversarial training flips a hidden relationship

A fingerprint of what robust training actually changes

- In an ordinary model, the images with the largest margin tend to have the *largest* gradients: confidence and sensitivity rise together.
- After adversarial training, this reverses: high-margin images now have the *smallest* gradients.
- The per-image relationship between margin and sensitivity flips sign, consistently across dozens of models.

Takeaway. Robust training does not just raise the margin. It re-couples margin and sensitivity, and η/L is the quantity that reads this change.

What it all means

1. **Stop using shift-consistency as a robustness signal.** It is saturated on modern models and does not order adversarial robustness.
2. **Distrust “invariance helps” claims tested with weak attacks.** They vanish under AutoAttack.
3. **Use the margin-to-sensitivity ratio η/L .** It predicts robustness without running an attack, from CNNs to frozen CLIP encoders.
4. **To compare strong models, look at the gradient’s shape,** not only its size. Concentrated sensitivity is more robust.

One idea, the margin relative to the sensitivity, unifies a split literature and turns a vague intuition about invariance into something you can measure and predict.

How to present this

The whole story in three beats

The three-beat story

- Beat 1. The confusion.** People treat invariance and robustness as the same thing and get contradictory results. They are not the same thing.
- Beat 2. The fix.** A simple ratio, margin over sensitivity, predicts robustness without any attack, and the field's invariance metric does not, because it is saturated.
- Beat 3. The refinement.** That ratio tells robust from fragile; to rank the robust ones you also need the shape of the model's sensitivity, and the apparent “invariance helps” effect is just a weak-attack mirage.

Takeaway. If you remember one line: *invariance is not robustness; the margin relative to the sensitivity is.*

Two analogies and the questions you will get

- **Analogy 1.** A locked door is not a strong door. Ignoring a gentle push (invariance) is not the same as withstanding a crowbar (an attack).
- **Analogy 2.** Two people equally far from a cliff edge: the one who moves slowly toward it is safer. That “distance over speed” is η/L .

Q: Isn't the ratio just re-describing the radius it predicts? Partly, and we say so; the surprise is that the invariance metric fails where the ratio succeeds.

Q: Does it hold on big models? Yes, on frozen CLIP encoders as well as CNNs.

Q: Can you optimize it directly? No, it is a diagnostic, not a training target; forcing it up collapses the margin.